05 MISSING DATA - CONCEPT + SYNTAX POSTER

IMPORTS

```
import numpy as np
import pandas as pd
from numpy import nan
```

FILL MISSING

```
ebola.fillna(0)
ebola = ebola.fillna(0)
ebola.ffill()
ebola.bfill()
ebola.interpolate()
ebola.interpolate(method=...)
```

NAN CHECKS

```
pd.isnull(nan)
pd.isnull(42)
pd.notnull(nan)
pd.notnull(42)
pd.isna(x)
pd.notna(x)
df.isna()
df.notna()
df.isnull()
df.notnull()
```

DO NOT USE

```
nan == nan
x == np.nan
df.fillna(method='ffill')
df.fillna(method='bfill')
inplace=True
.ix[...]
```

READ CSV NA

```
pd.read_csv('survey_visited.csv')
pd.read_csv('survey_visited.csv',
   keep_default_na=False)
pd.read_csv('survey_visited.csv', na_values=[''],
   keep_default_na=False)
pd.read_csv('file.csv', na_values=[99])
keep_default_na=False
na_filter=False
```

DROP MISSING

```
ebola.dropna()
ebola.dropna(how='any')
ebola.dropna(how='all')
ebola.dropna(thresh=10)
ebola.dropna().shape
```

CREATE NAN

```
pd.Series({'goat': 4, 'amoeba': np.nan})
scientists['missing'] = np.nan
```

REINDEX NAN

```
life_exp = gapminder.groupby('year')['lifeExp'].mean()
y2000 = life_exp[life_exp.index > 2000]
y2000.reindex(range(2000, 2010))
```

CALCULATIONS

```
ebola.sum()
ebola.sum(skipna=True)
ebola.sum(skipna=False)
ebola.mean()
ebola.count()
ebola.min()
ebola.max()
```

MERGE NAN

```
visited.merge(survey, left_on='ident',
   right_on='taken')
```

COUNT MISSING

```
ebola.shape
ebola.count()
ebola.shape[0] - ebola.count()
np.count_nonzero(ebola.isnull())
np.count_nonzero(ebola['Cases_Guinea'].isnull())
ebola['Cases_Guinea'].value_counts(dropna=False).head()
df.isnull().sum()
df.isna().sum()
```